RED FM

Weekly checks

Border Holdings, Avonmouth

Date: 2026-10-14

Ref:	WK-MV83YSEQ
Date:	2026-10-14
Engineer:	Dexter Norley
Plant:	A & B, C & D, E
Week:	42
Ambient °C:	15
Faults recorded:	0

Faults recorded on this visit

None - no line item was flagged.

A & B - Chamber temperature

Item	A1	A2	A3	A4	No1 B2	No2 B2	No3 B1	No4 B1
Cold store set point (controller)	-20	-20	-20	-20	-18.9	-18.9	-18.9	-18.9
Cold store temp air-on	-19.8	-19.9	-20.4	-19.9	-18.6	-18.7	-18.8	-19.1
Coolers air off	-21.6	-21.7	-22.8	-21.5	-20.3	-19.7	-19.9	-20.9
Cooling on/off	On	On	On	0n	On	On	On	On

A & B - Coolers

Item	A1	A2	A3	A4	No1 B2	No2 B2	No3 B1	No4 B1
Cooler fan correct operation	OK	OK	OK	OK	OK	OK	OK	OK
Cooler correct operation	OK	OK	OK	OK	OK	OK	OK	OK
Cooler correct defrosting cycle	OK	OK	OK	OK	OK	OK	OK	OK
Cooler ice build-up & debris	OK	OK	OK	OK	OK	OK	OK	OK
Defrost drain line for leakage	OK	OK	OK	OK	OK	OK	OK	OK
Operating suction superheat #	6c	6.8c	9c	8.8c	6.8c	6.7c	7.1c	7c
% opening of expansion valve #	26%	32%	64%	54%	53%	14%	21%	19%
Cooler air temp diff #	1.8	1.8	2.4	1.6	1.7	1	1.1	1.8

A & B - Compressor racks & condenser

Item	A	B
Compr abnormal noise / vibration	OK	OK
Oil cooler fin face free of debris	OK	OK
Oil cooler fans correct operation	OK	OK
Plant-room ventilation fan (on/off)	On	On
Plant-room temperature	30	30
Wet-matt cooling on/off (hours)	On	Off
System delivery pressure	18bar	15.1
Condenser fin face free of debris	OK	OK
Condenser fans correct operation	Fan 3 u/s	All working
Liquid line sight glass	Clear	Clear
Receiver level	Above bottom sight glass	Below bottom sight glass
Oil separator oil level	Full	Full
No1 compressor running & % load	100%	Off
No2 compressor running & % load	75%	100%

Item	A	B
No3 compressor running & % load	Off	-
No1 compr delivery & econ temp	Delivery 68c econ 9c	Off
No2 compr delivery & econ temp	Delivery 66c econ off	Delivery 65c econ 5c
No3 compr delivery & econ temp	Comp is off	-
Refrigerant leak detection system	OK	OK

A & B - Controls

Item	A	B
Suction pressure	0.8bar	0.9
Suction temperature	-17	-18.7
Suction SH	10c	7c
Delivery pressure	17.9bar	15.1
Common delivery temp	75.5c	72.4
Oil temperature	45c	46.2
HP liquid temperature	36c	31c
HP liquid sub-cooling	7c	6c
Econ liquid temperature	21c	1c
Econ sub-cooling	15c	30c

A & B - Main incoming meter

Item	kWhr
Electric kWhr meter reading	707078
Previous meter reading	701783
Total kWhr	5295

A & B - A - pack meter

Item	kWhr
Electric kWhr meter reading	4664955
Previous meter reading	4652519
Pack total kWhr	12436
A chamber total energy	16470

A & B - B - pack meter

Item	kWhr
Electric kWhr meter reading	2886598
Previous meter reading	2876761
Pack total kWhr	9837
B chamber total energy	13639

A & B - A & B cooler energy meter

Item	kWhr
Electric kWhr meter reading	2657286
Previous meter reading	2649450
Total kWhr	7836

A & B - B cooler energy meter

Item	kWhr
Electric kWhr meter reading	717752
Previous meter reading	713950
Total kWhr	3802

A & B - A cooler energy (derived)

Item	kWhr
Electric kWhr meter reading	1939534
Previous meter reading	1935500
Total kWhr	4034

C & D - Chamber temperature

Item	C1	C2	C3	C4	D1	D2
Cold store set point (controller)	-18.8	-18.8	-18.8	-18.8	-18.6	-18.1
Cold store temp air-on	-18.2	-18.7	-18.8	-19.1	-18.8	-18.5
Coolers air off	-21.9	-19.8	-19.9	-20.2	-20.9	-20
Cooling on/off	On	On	On	Off	On	On

C & D - Condensing units

Item	C1	C2	C3	C4	D1	D2	D3	D4
Compr abnormal noise / vibration	OK	OK	OK	OK	OK	OK	OK	OK
Compressor abnormal frosting	OK	OK	OK	OK	OK	OK	OK	OK
Compr oil level	2/3	2/3	2/3	2/3	2/3	2/3	2/3	2/3
Compr delivery temperature	71	69	72	73	75	76	75	75
Oil recovery / separator level	Full	Full	Full	Full	Full	Full	Full	Full
Condenser fin face free of debris	OK	OK	OK	OK	OK	OK	OK	OK
Condenser fans abnormal noise	OK	OK	OK	OK	OK	OK	OK	OK
Liquid line sight glass	OK	OK	OK	OK	OK	OK	OK	OK
Receiver level	Below bottom sight glass	Above bottom sight glass	Below bottom sight glass	Above bottom sight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass	Below bottom sight glass

C & D - Coolers

Item	C1	C2	C3	C4	D1	D2
Cooler temperature diff	3.7	1.1	1.1	1.1	2.1	1.5
Cooler correct operation	OK	OK	OK	OK	OK	OK
Cooler correct defrosting cycle	OK	OK	OK	OK	OK	OK
Cooler ice build-up & debris	OK	OK	OK	OK	OK	OK
Defrost drain line for leakage	OK	OK	OK	OK	OK	OK

C & D - Energy readings

Item	C2-4 refrig	C1-3 refrig	D 1&2	D 3&4 compr
Electric kWhr meter reading	3030717	1136407	2658058	24572
Previous kWhr meter reading	3026377	1133153	2651107	24529
Total kWhr	4340	3254	6951	43
Chamber total energy		7594		6994

E - Chamber temperature

Item	E1	E2
Cold store set point (controller)	-18.6	-18.6
Cold store temp air-on	-18.7	-18.8
Coolers air off	-19.4	-19.7
Cooling on/off	On	On

E - E chamber front-centre temp

Item	°C
Front-centre temperature	-18.6

E - Condensing units

Item	E1 Compr1	E1 Compr2	E2 Compr1	E2 Compr2
Compr abnormal noise / vibration	OK	OK	OK	OK
Compressor abnormal frosting	OK	OK	OK	OK
Compr oil level	2/3	2/3	2/3	2/3
Compr delivery temperature	64	65	66	65
Oil recovery / separator level	Full	Full	Full	Full
Condenser fin face free of debris	OK	OK	OK	OK
Condenser fans abnormal noise	OK	OK	OK	OK
Liquid line sight glass	OK	OK	OK	OK
Receiver level	Above bottom sight glass	Above bottom sight glass	Below bottom sight glass	Below bottom sight glass

E - Coolers

Item	E1	E2
Cooler temperature diff	0.7	0.9
Cooler correct operation	OK	OK
Cooler correct defrosting cycle	OK	OK
Cooler ice build-up & debris	OK	OK
Defrost drain line for leakage	OK	OK

E - System readings

Item	E1	E2
Suction pressure	0.6	0.7
Suction temperature	-20	-19
Suction superheat	9c	9c
Delivery pressure	13.7	14
Liquid refrig temperature	31	32
Liquid sub	4c	4c

E - Energy readings

Item	E1	E2
Electric kWhr meter reading	14374	12731
Previous kWhr meter reading	14308	12700
Total kWhr	66	31
Chamber total energy		97

RDM alarms at time of visit

Item	Asset / chamber	Active since	Outcome	Action taken / reason
S02 stage fault	A plant	2026-09-30	Ongoing	New condenser on order.